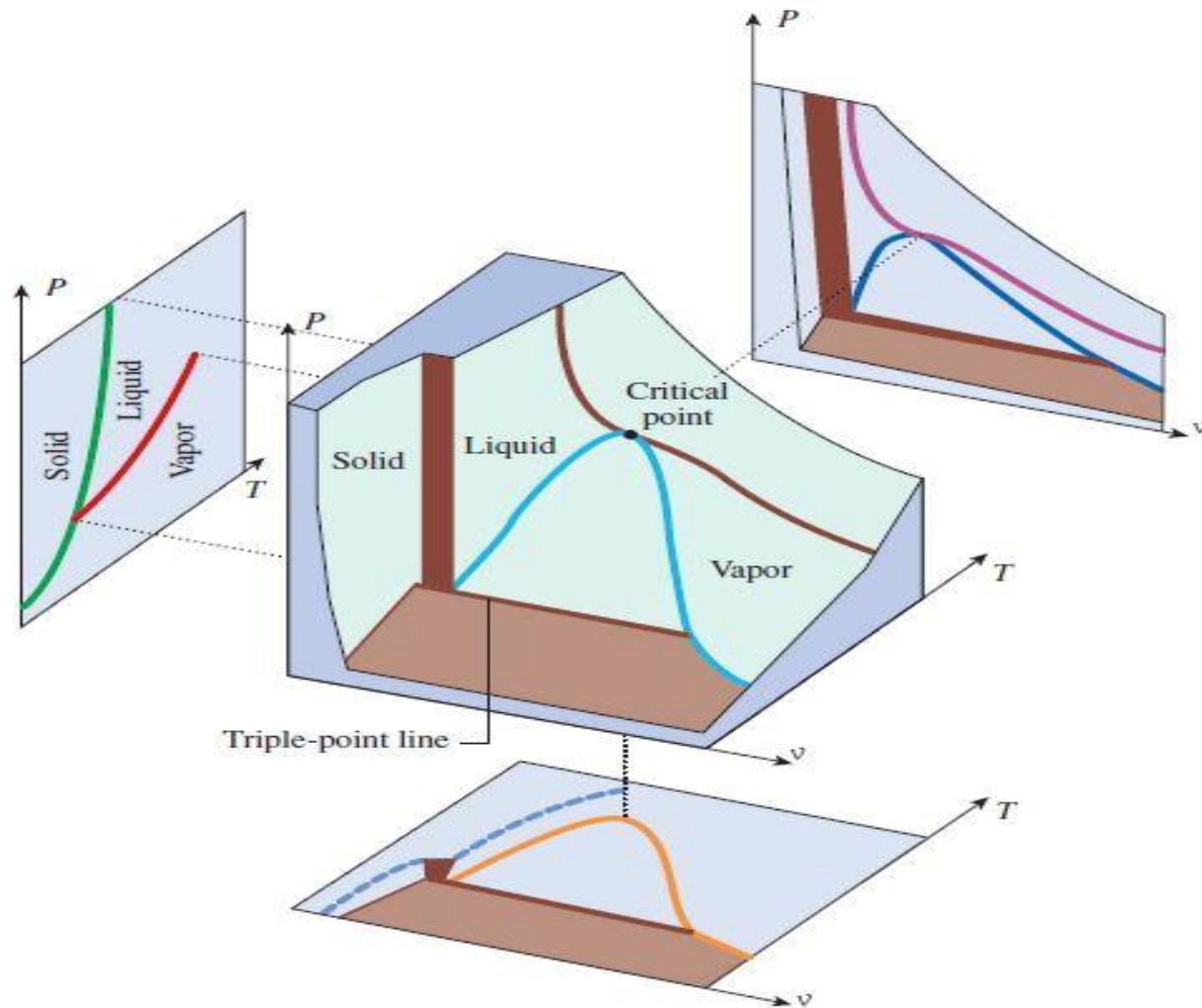




Module-2: Properties of Pure Substance

Thermodynamics (NMEC103) : (Winter 2024-25)

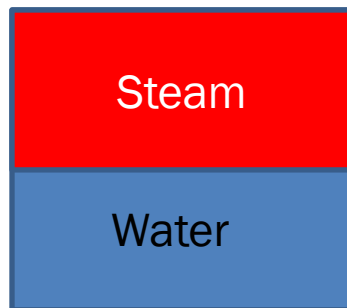
Dr. S. Sahoo, Mechanical Engineering, IIT Dhanbad

Properties Of Pure Substance

Any system uses a working substance (may be pure substance /multi component mixture etc.) for energy transfer or transformation.

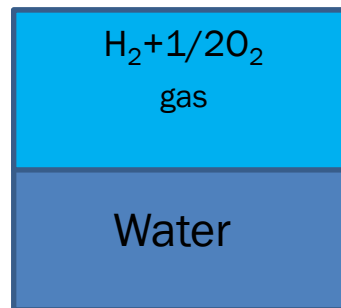
Pure substance:

1. Homogeneous in chemical composition
2. Invariable in chemical composition
3. Homogeneous in chemical aggregation
4. Invariable in chemical aggregation
5. It may have more than one phase



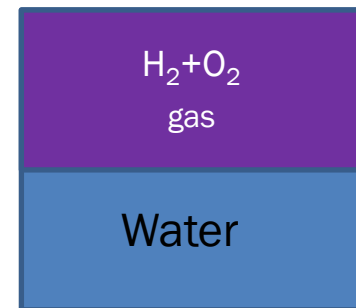
1&2

System-1



1

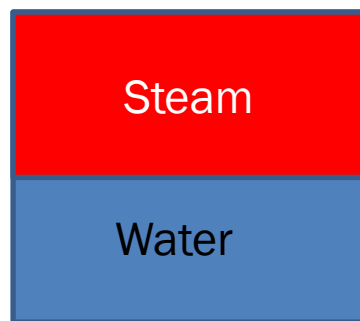
System-2



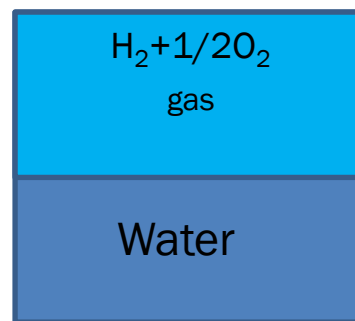
None

System-3

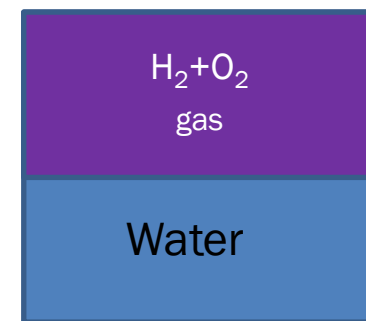
“Homogeneous in composition” means that the composition of each part of the system is the same as the composition of every other part. “Composition” means the relative proportions of the chemical elements into which the sample can be analyzed. It does not matter how these elements are combined. As shown in system 1 both steam and water have the same composition ($\text{H}:\text{O}=2:1$), similarly in system-2 both the gas mixture and water have the same composition ($\text{H}:\text{O}=2:1$), however in system-3 the composition is inhomogeneous gas mixture ($\text{H}:\text{O}=1:1$), Water ($\text{H}:\text{O}=2:1$),



System-1



System-2



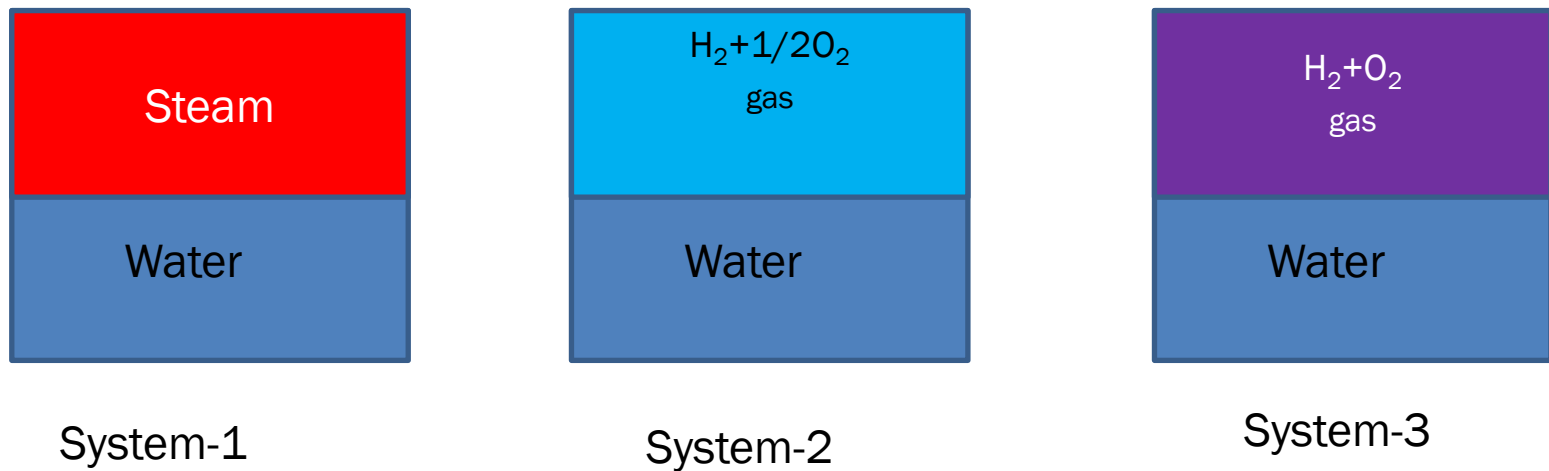
System-3

Both the system-1 & 2 satisfy the condition 1&2 but System-3 does not

The system-1 satisfies condition 3&4 but system2 &3 do not, hence System-1 is pure substance

“Homogeneous in chemical aggregation” means that the chemical elements must be combined chemically in the same way in all parts of the system

As shown in system-1 both steam and water have the same Chemical aggregation i.e. Hydrogen and Oxygen are bonded similarly both in steam and water, However in system-2 and 3 the bonding between Hydrogen and Oxygen in gas mixture and water are different

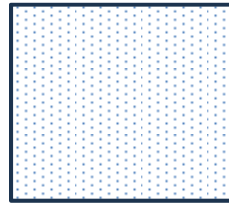


“In Variable in chemical aggregation ” means the state of the chemical combination of the system does not change with time

Homogeneous (Single Phase)

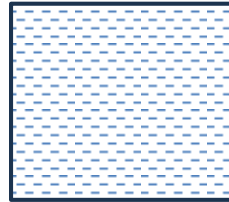
Heterogeneous (multiphase)

Singlecomponent



Pure substance

Steam (H_2O)

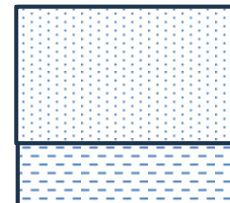


Water (H_2O)

Simple Substance

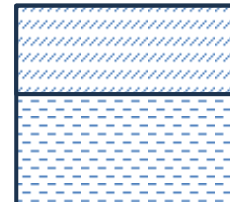


Ice (H_2O)

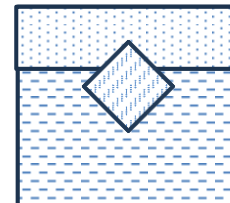


Pure substance

Steam & water

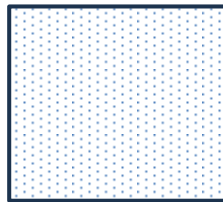


Ice & water



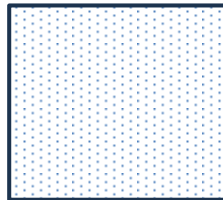
Steam, Water and Ice

Multicomponent

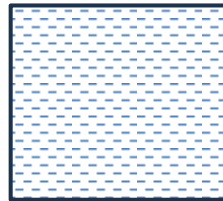


Pure substance

Air (Gaseous mixture)
(N_2 , O_2 , CO_2 etc.)

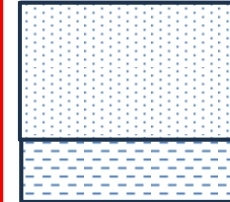


Product of Combustion
gaseous mixture: CO_2 ,
 H_2O , N_2 , O_2)



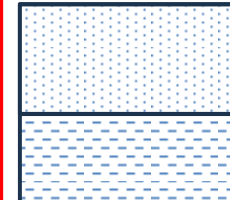
Sea water (Liquid
mixture: H_2O , NaCl ,
etc.)

Simple Substance

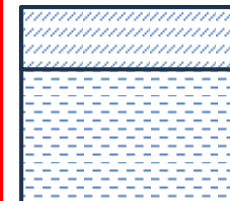


Not a Pure substance

Gaseous air & Liquid Air



Gaseous mixture
(fuel vapor, air)
and liquid fuel



Sea ice and
sea water

1. Is moist air a pure substance ?

It is well known that moist air is a homogeneous mixture of different **permanent gases** (O_2 , N_2 , CO_2 , Ar, etc.) and **water vapour**

Dry air :

Constituent	Molecular weight	Mol fraction
Oxygen	32.000	0.2095
Nitrogen	28.016	0.7809
Argon	39.944	0.0093
Carbon dioxide	44.010	0.0003

The **dry air** & **water vapour** both are **pure substances**, however the water vapour part of moist air may change in any air conditioning process, hence **moist air is not a pure substance**, rather a **binary mixture** of dry air and water vapour.



- Moist air is less dense compared to the dry air. $M_{H_2O} = 18 \text{ kg/kmol}$

2. Is mixture of liquid air a gaseous air pure substance ?

- At standard atmospheric pressure and temperature, the dry air is a mixture of approximately 78.09% N_2 , 20.95% O_2 , 0.93% Ar, 0.04% CO_2 , and other gases in meagre amounts.
- The **gaseous** as well as **liquid air** are pure substances, however, the **mixture of gaseous and liquid air is not a pure substance**. The concentration of N_2 in liquid air is higher than that in the gaseous air, then the chemical composition is no more homogeneous.



3. Is mixture of ice, water and steam is a pure substance ?

As the chemical composition as well as aggregation in all these three phases of water is same, the mixture can be treated as pure substance

4. Is combustion products is a substance ?

In a combustion process the reactants (air and fuel mixture) is not a pure substance but the combustion product (homogeneous mixture of CO_2 NO_x CH_4 etc.) can be treated as a pure substance

Important characteristics of pure substance:

State Postulate: To specify the state of a simple compressible pure substance two independent intensive properties are required. Simple system means absence of the effect of electric, magnetic, anisotropic stress field, surface tension, gravity etc. There is only one mode of work interaction. If it is a compressible system that one mode is the moving boundary work. The number 2 came from the fact that 1 related to heat interaction and 1 is related to the single mode of work interaction.

One of the important property of a pure substance is that during the phase change process the pressure and temperature are dependent properties. So if you fix the pressure the temperature is automatically fixed during phase change (water boils at 100°C when the pressure is maintained at 1 bar). The azeotropic mixtures also behave like pure substance.

Pure Substance						Pure substance
	1	2	3	4	5	
Steam	✓	✓	✓	✓	1	✓
Steam & water	✓	✓	✓	✓	2	✓
Air (O ₂ +N ₂ gas)	✓	✓	✓	✓	1	✓
Air + liquid air	✗	✗	✓	✓	2	✗
Air + fuel vapor no reaction	✓	✓	✓	✓	1	✓
Air +fuel vapor & liquid fuel	✗	✗	✓	✓	2	✗
Air + fuel vapor & combustion product	✓	✓	✗	✗	2	✗
Gaseous Combustion Products	✓	✓	✓	✓	1	✓

Gibb's Phase Rule (1876)

To identify the degrees of freedom (No of independent intensive properties) of multi component and multiphase systems, the state postulate is handicapped. In such cases Gibbs phase rule is useful. It is still valid only for simple systems.

The component C of the system are the chemical species from which the system, in any of its states, can be prepared. However there is a caveat: To be considered as components, the various substance must be capable of independent variation. Thus, the number of components of a system is the least number of independently variable, chemically distinct species in the system. In complex system, determining the number of component requires careful analysis because the number of components is not always equal to the number of different species present.

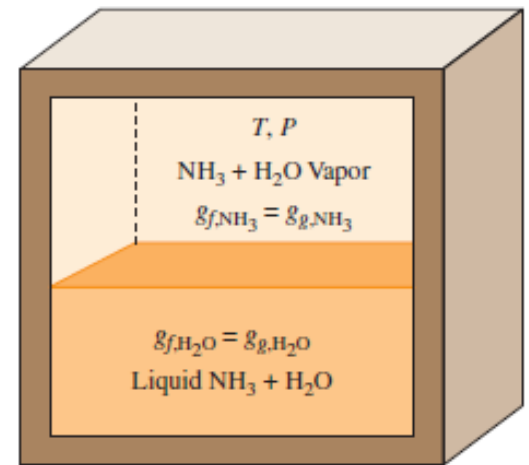
The Phase Rule. The phase rule was 1st published by J. Willard Gibbs in 1876. It is used to determine the maximum number of intensive properties or degrees of freedom for a system in equilibrium. This number of property is necessary and sufficient to specify the state of each phase in the system, but it is insufficient to specify the amount of each phase present . In other words the number properties specified by the Gibb's phase rule is sufficient to determine the intensive state but not the extensive state of the system

For any system in equilibrium that is homogeneous (or composed of a finite number of homogeneous parts in contact) and free from electric, magnetic, gravitational, solid distortion, and surface tension effects, the phase rule states that the maximum number of degrees of freedom is given by:

let us consider a **heterogeneous system** consisting of **P number of phases** and **C number of components** in **equilibrium**. Let us assume that **the passage of a component from one phase to another doesn't involve any chemical reaction**. When the system is in equilibrium, it can be described by the following parameters

The total number of variables required to specify the state of the system is:

- **pressure same for all phases**
- **Temperature same for all phases**
- **Concentration**



The independent concentration variable for one phase with respect to the C components is **$C - 1$**

Therefore, the independent concentration variables for P phases with respect to C components is **$P (C - 1)$** .

Total number of variables = $P (C - 1) + 2 \dots (1)$

The total number of equilibria

The various phases present in the system can only remain in equilibrium when the chemical potential (μ) of each of the component is the same in all phases, i.e.

The condition of phase equilibrium at particular pressure and temperature yields

$$\mu_1, P_1 = \mu_1, P_2 = \mu_1, P_3 = \mu_1, P_4 \dots \dots \dots \mu_1, P$$

$$\mu_2, P_1 = \mu_2, P_2 = \mu_2, P_3 = \mu_2, P_4 \dots \dots \dots \mu_2, P$$

$$\mu_3, P_1 = \mu_3, P_2 = \mu_3, P_3 = \mu_3, P_4 \dots \dots \dots \mu_n, P$$

$$\mu_c, P_1 = \mu_c, P_2 = \mu_c, P_3 = \mu_c, P_4 \dots \dots \dots \mu_c, P$$

The number of equilibria for *each P phases for each component* is **$P - 1$** .

Hence, the total number of equilibria involved is

$$\mathbf{E = C (P - 1) \dots (2)}$$

Equating eq (1) and (2), we get

$$F = P(C - 1) + 2 - C(P - 1)$$

$$F = CP - P + 2 - CP + C$$

$$F = C - P + 2$$

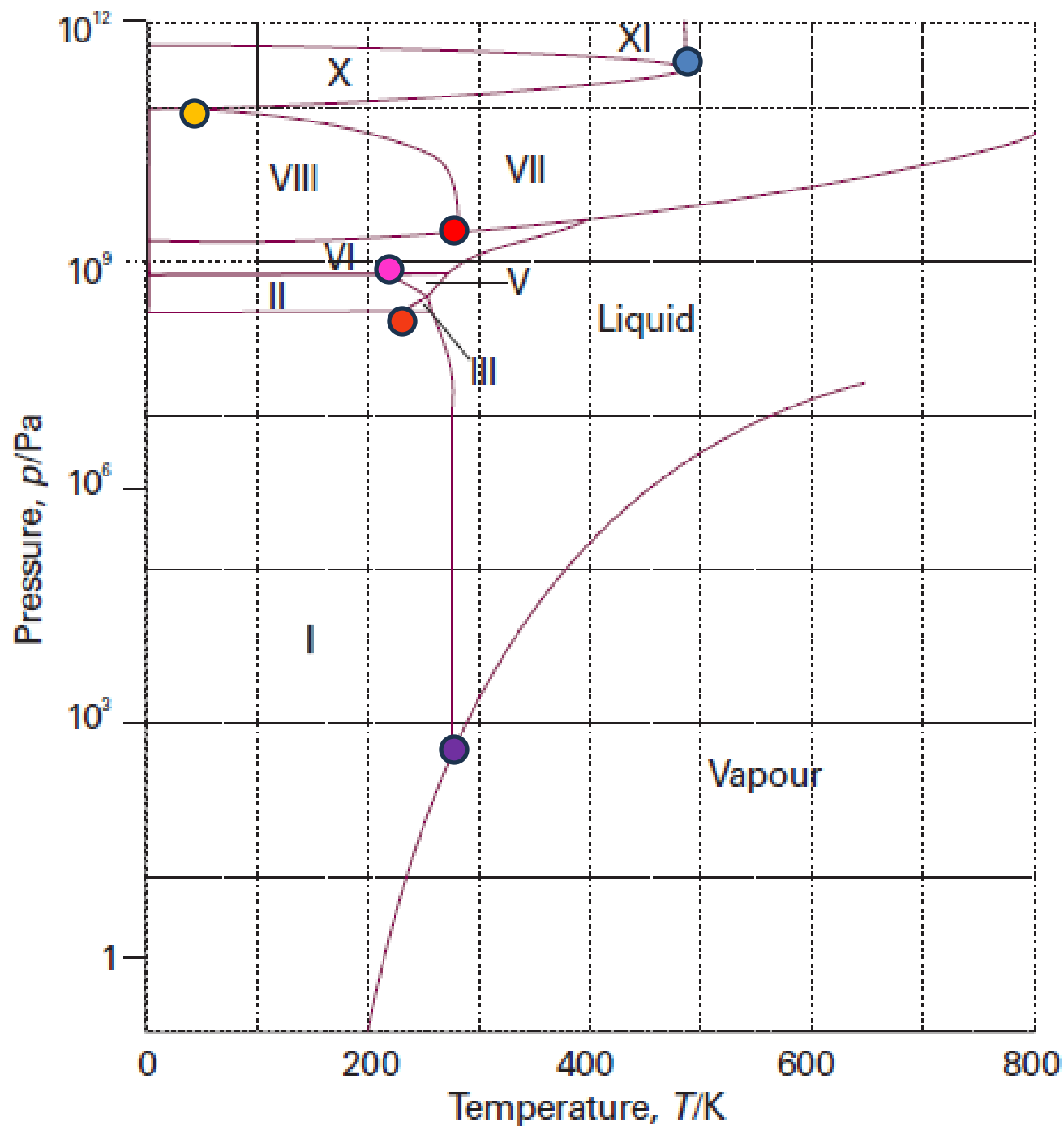
$$F = C + 2 - P$$

$F \rightarrow$ Degrees of freedom (No of independent intensive property)

$P \rightarrow$ Number of phases

$C \rightarrow$ Number of Components (May not be equal to number of chemical species)

The minimum allowable value for the degrees of freedom is zero: $F = 0$. For a single component system the Gibbs phase rule corresponds to $P = 3$ a three phase system. Thus 3 is the maximum number of different phases of pure component that can coexist in equilibrium. For ice, liquid water and water vapor to be in equilibrium the temperature must be 0.01°C and pressure must be 0.6113 kPa



e.g.1 Moist air: Number of Components - 2 (dry air and water vapour)

Number of Phases -1 (all are in gaseous phase)

$$\text{So } F = C + 2 - P = 2 + 2 - 1 = 3$$

So to fix the intensive state of moist air we have specify at least three independent intensive property out of which one is the barometric pressure.

e.g.2 Water, Ice and steam are in equilibrium at the triple point:

Number of Components 1 (H_2O)(chemical composition i.e. H:O= 2:1 in all phases)

Number of Phases -3 (Solid, Liquid, Vapour)

$$\text{So } F = C + 2 - P = 1 + 2 - 3 = 0$$

So at triple point there are no independent variable , so degrees of freedom is zero (the system is invariant) i.e. if you vary any one of the intensive properties the equilibrium of three phases will not exist

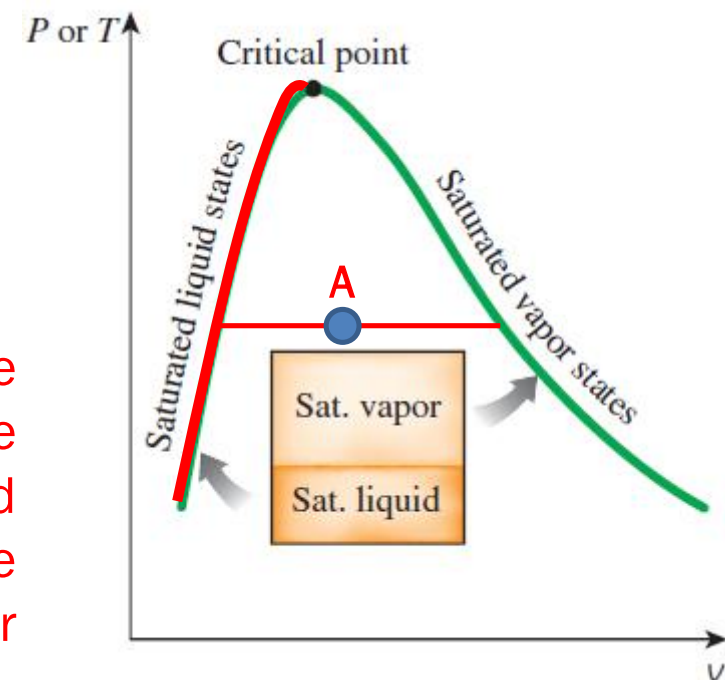
e.g.3 Consider a system of ammonia and water in liquid solution that is in equilibrium with a vapour, Determine the number of degrees of freedom for this binary mixture: There are two phases , liquid and vapour, two components NH_3 & H_2O , or NH_4OH and H_2O . From the phase rule $F = C + 2 - P = 2 + 2 - 2 = 2$

This means that specifying the pressure and temperature of this system completely determine the composition of each phase

e.g. 4 Consider a system a mixture of saturated liquid and saturated vapor as shown by state A in the figure

$$F = C + 2 - P = 1 + 2 - 2 = 1$$

Only one property either pressure or temperature is enough to specify the intensive state as the liquid is saturated liquid and vapor is at saturated vapor at that pressure. But average property of the mixture can be obtained only when another property is specified (x)



T-V diagram for a pure substance

Consider as a system 1 kg of water contained in the piston/cylinder arrangement shown in Fig. 1a. Suppose that the piston and weight maintain a **pressure of 0.1 MPa** in the cylinder and that the initial **temperature is 20°C**. As **heat is transferred to the water**, the **temperature increases appreciably**, the **specific volume increases slightly**, and the **pressure remains constant**. When the **temperature reaches 99.6°C**, additional heat transfer results in a change of phase, as indicated in Fig. 1b. That is, some of the liquid becomes vapor, and during this process **both the temperature and pressure remain constant**, but the **specific volume increases considerably**. When the last drop of liquid has vaporized, further transfer of heat results in an increase in both the temperature and specific volume of the vapor, as shown in Fig. 1c.

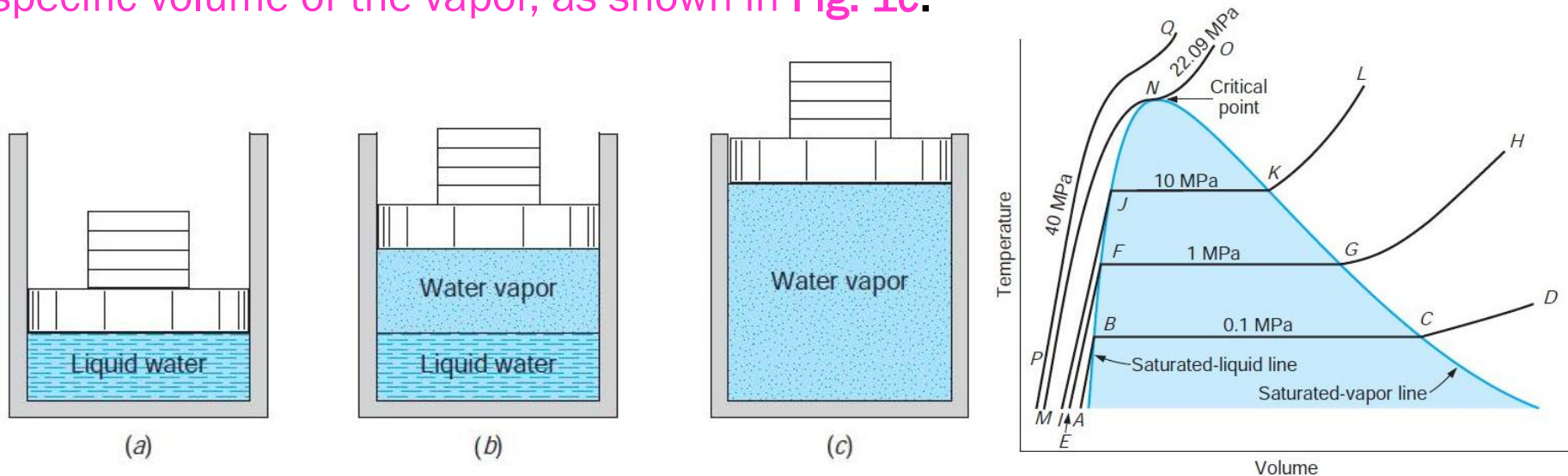
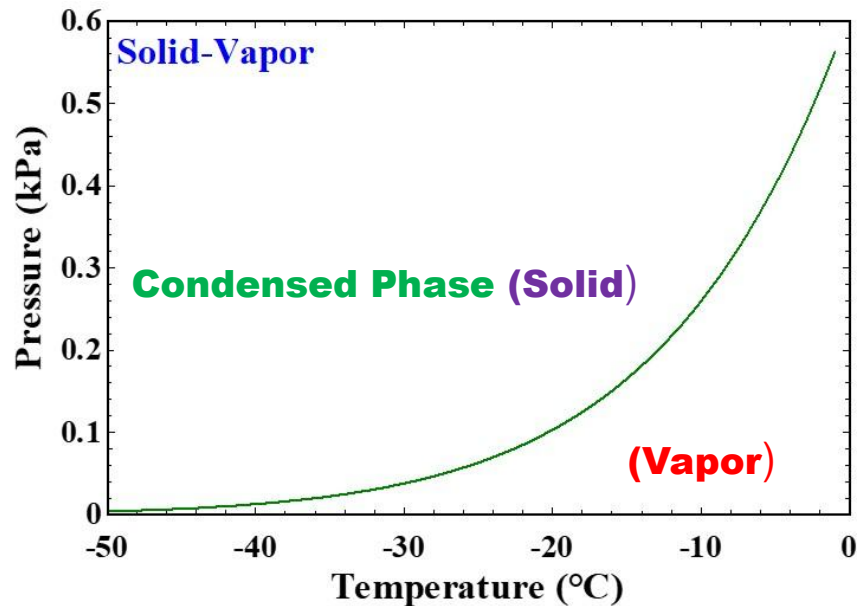
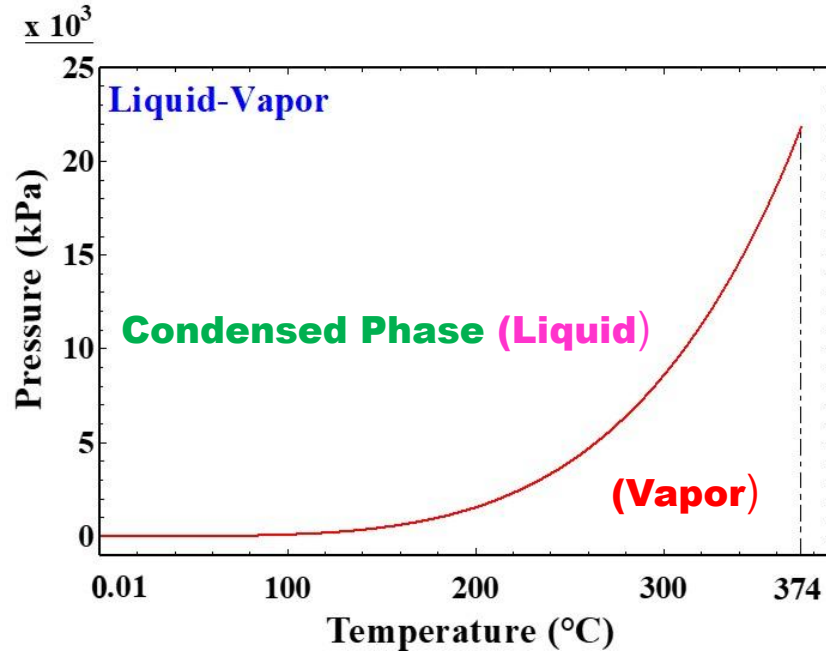
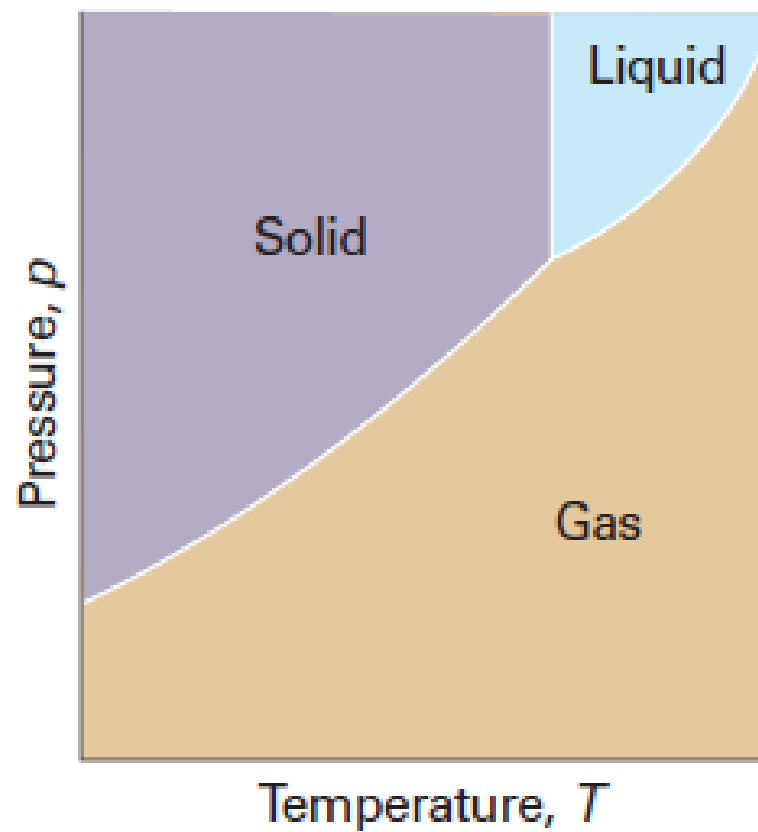
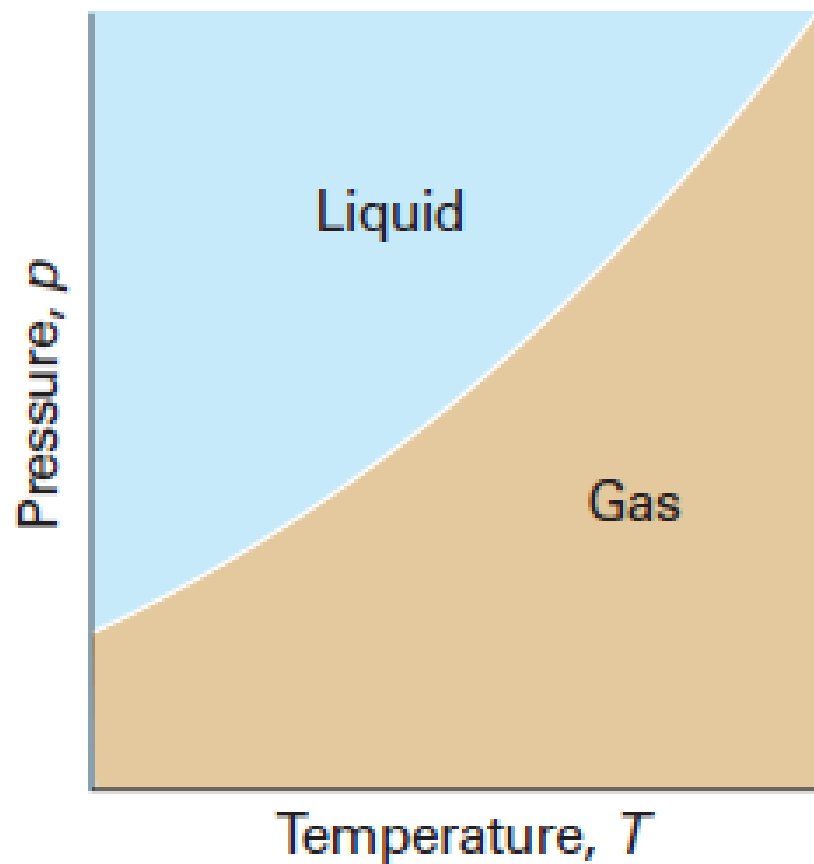


Fig.1

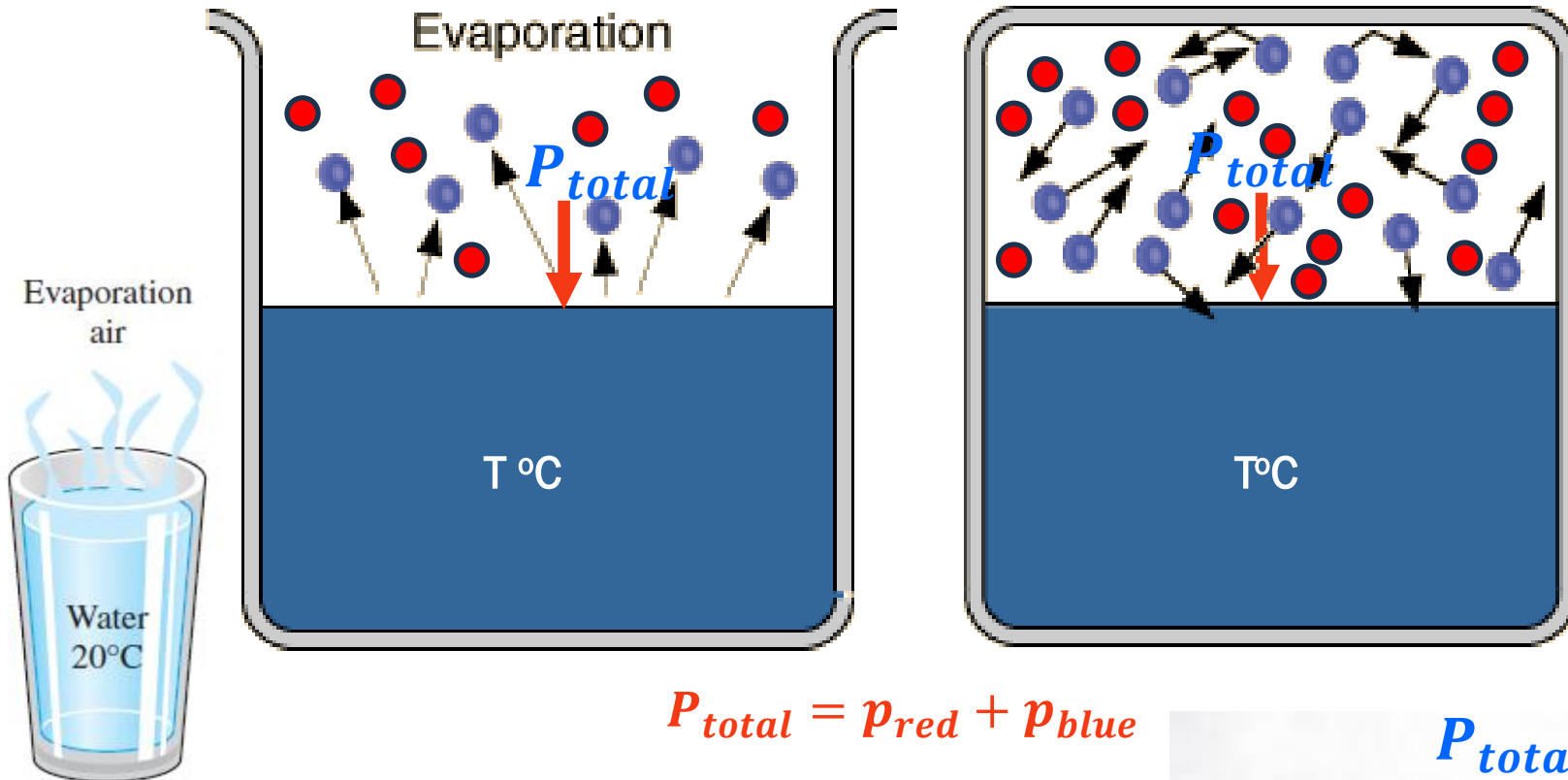
The term **saturation temperature** designates the temperature at which vaporization takes place at a given pressure. This pressure is called the **saturation pressure** for the given temperature. Thus, for water at **99.6°C** the saturation pressure is **0.1 MPa**, and for water at **0.1 MPa** the saturation temperature is **99.6°C**. For a pure substance there is a definite relation between saturation pressure and saturation temperature. A typical curve, called the *vapor-pressure curve*, is shown in the figure



If a substance exists as liquid at the saturation temperature and pressure, it is called a **saturated liquid**. If the temperature of the liquid is lower than the saturation temperature for the existing pressure, it is called either a **subcooled liquid** (implying that the temperature is lower than the saturation temperature for the given pressure) or a **compressed liquid** (implying that the pressure is greater than the saturation pressure for the given temperature).



Vapor Pressure



$$P_{total} = p_{red} + p_{blue}$$

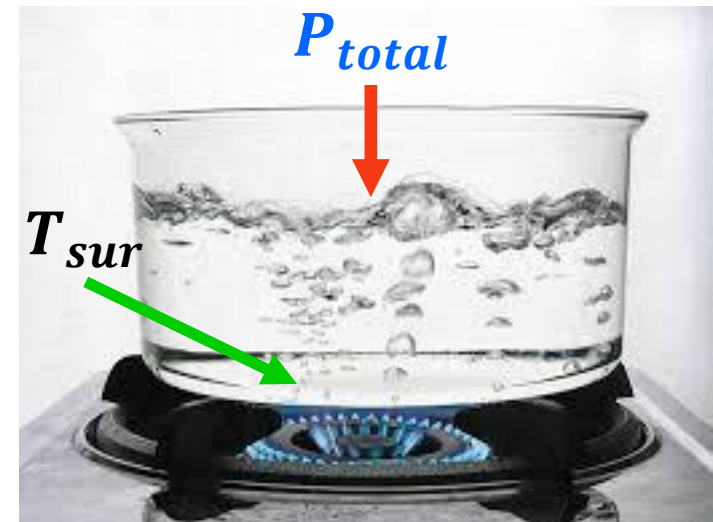
$$p_{vap} = p_{blue}$$

At equilibrium: $p_{vap} = p_{sat}(T)$

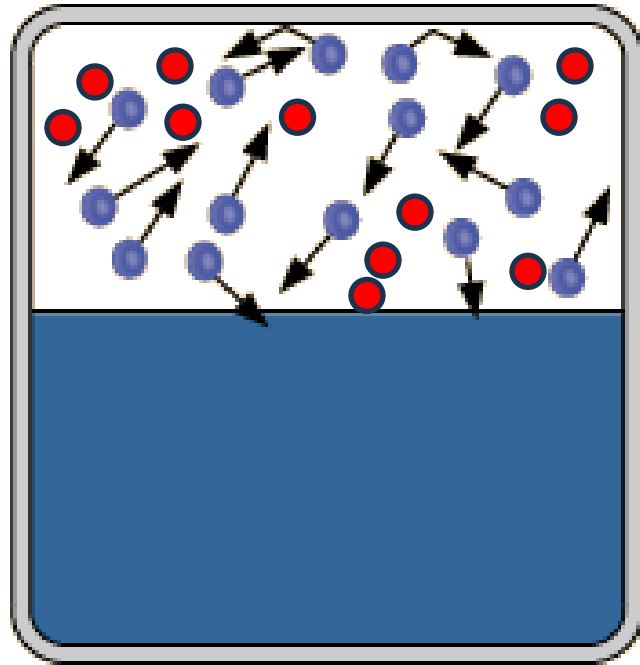
Evaporation occurs when: $p_{blue} = p_{vap} < p_{sat}(T\text{ }^{\circ}\text{C})$

Boiling occurs when: $p_{sat}(T\text{ }^{\circ}\text{C}) \geq P_{total}$

Boiling occurs when: $T_{sur} > T_{sat}(P_{total})$



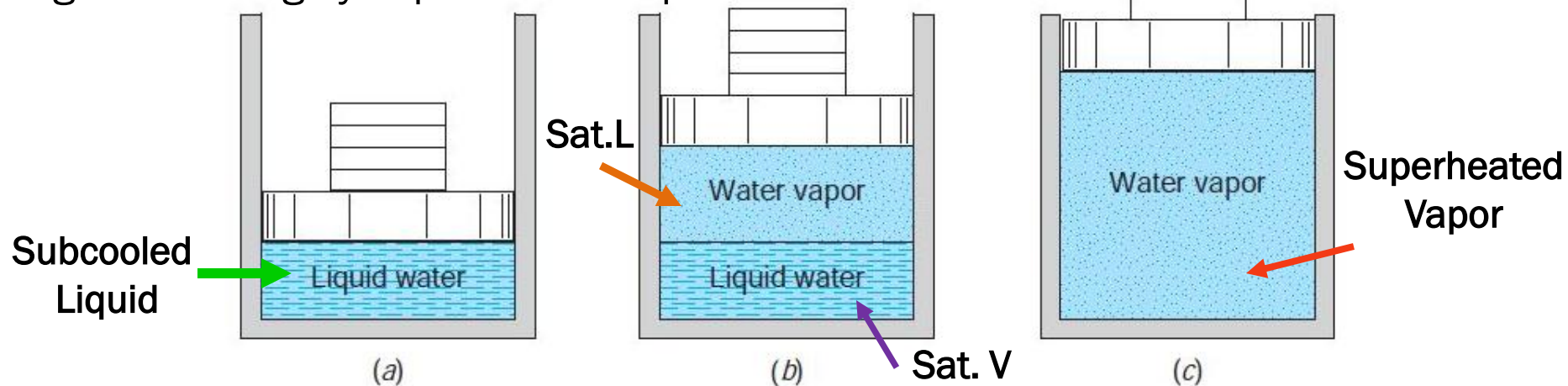
When a liquid is **exposed to a gaseous medium** some of the liquid particles from the free surface **evaporates** and **stays in gaseous phase**. **At equilibrium** the **rate of vaporisation becomes equal to the rate of condensation** and the partial pressure of this vapor is called vapor pressure and is equal to the saturation pressure corresponding to its temperature. But actually the partial pressure exerted by the vapor of the liquid is slightly lower than the saturation pressure due to the presence of the other gases which affects the escaping and condensing tendency of the vapor particle of the liquid.



$$p_{vap} = p_{sat}(T) \text{ (from steam table)}$$

$$p_{vap} \approx p_{sat}(T) \text{ (From steam table)}$$

- When a substance exists as part liquid and part vapor at the saturation temperature, its **quality** is defined as the ratio of the mass of vapor to the total mass. Thus, in Fig. 1b, if the mass of the vapor is 0.2 kg and the mass of the liquid is 0.8 kg, the quality/dryness fraction is 0.2 or 20%. The quality/dryness fraction may be considered an **intensive property** and has the symbol x . Quality has meaning only when the substance is in a saturated state, that is, at saturation pressure and temperature. If a substance exists as vapor at the saturation temperature, it is called **saturated vapor**. (Sometimes the term *dry saturated vapor* is used to emphasize that the quality is 100%.) $x = m_v / (m_v + m_L)$ or $m_g / (m_g + m_f)$
- When the vapor is at a temperature greater than the saturation temperature, it is said to exist as **superheated vapor**. The pressure and temperature of superheated vapor are independent properties, since the temperature may increase while the pressure remains constant. Actually, the substances we call gases are highly superheated vapors.



Consider Fig. 1 again. Let us plot on the temperature-volume diagram of Fig. 2 the constant-pressure line that represents the states through which the water passes as it is heated from the initial state of 0.1 MPa and 20°C . Let state A represent the initial state, B the saturated-liquid state (99.6°C), and line AB the process in which the liquid is heated from the initial temperature to the saturation temperature. Point C is the saturated-vapor state, and line BC is the constant-temperature process in which the change of phase from liquid to vapor occurs. Line CD represents the process in which the steam is superheated at constant pressure. Temperature and volume both increase during this process.

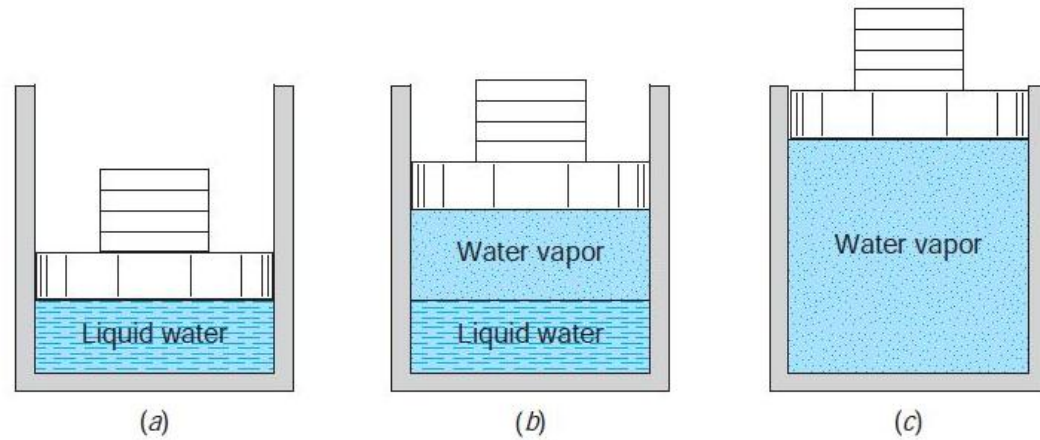


Fig. 1

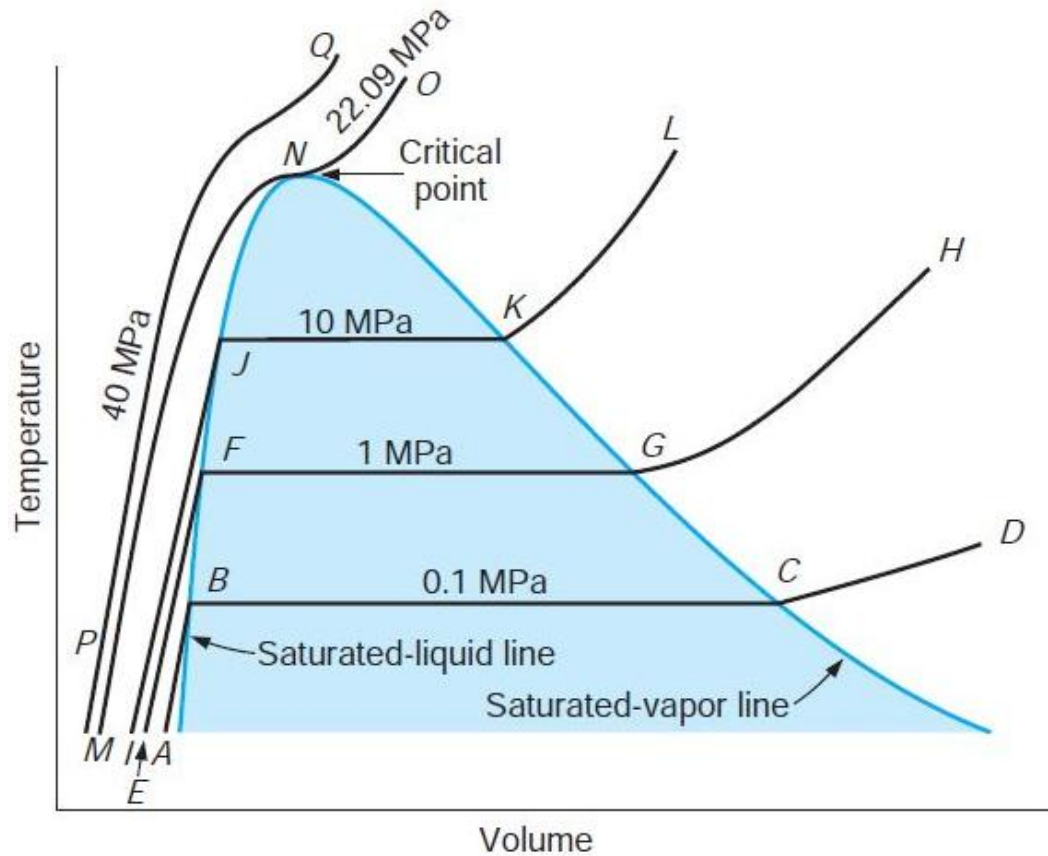
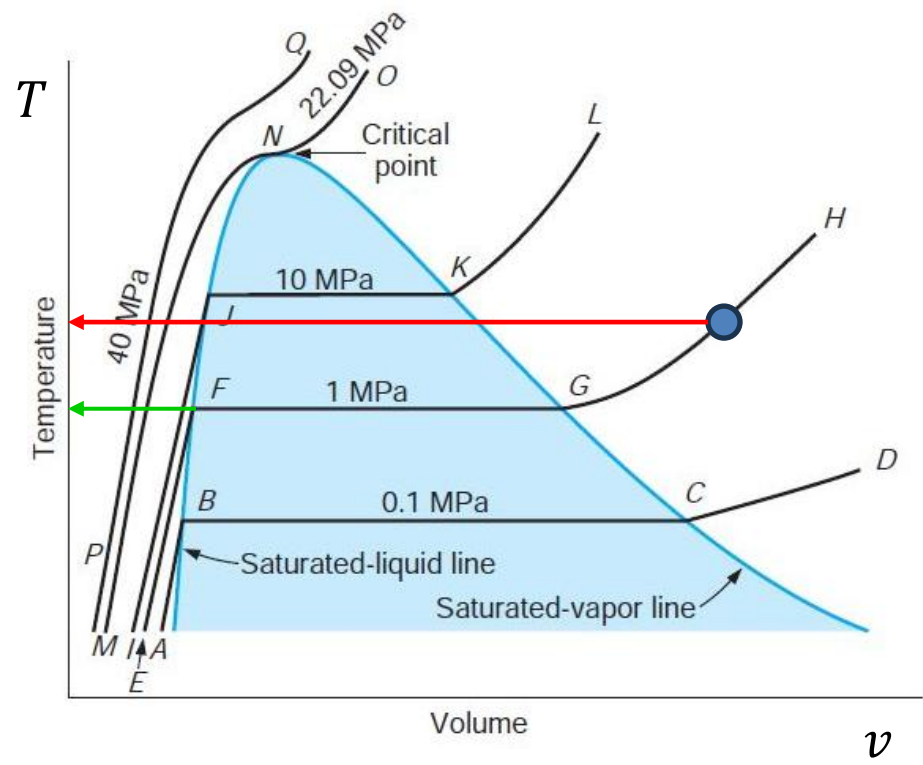
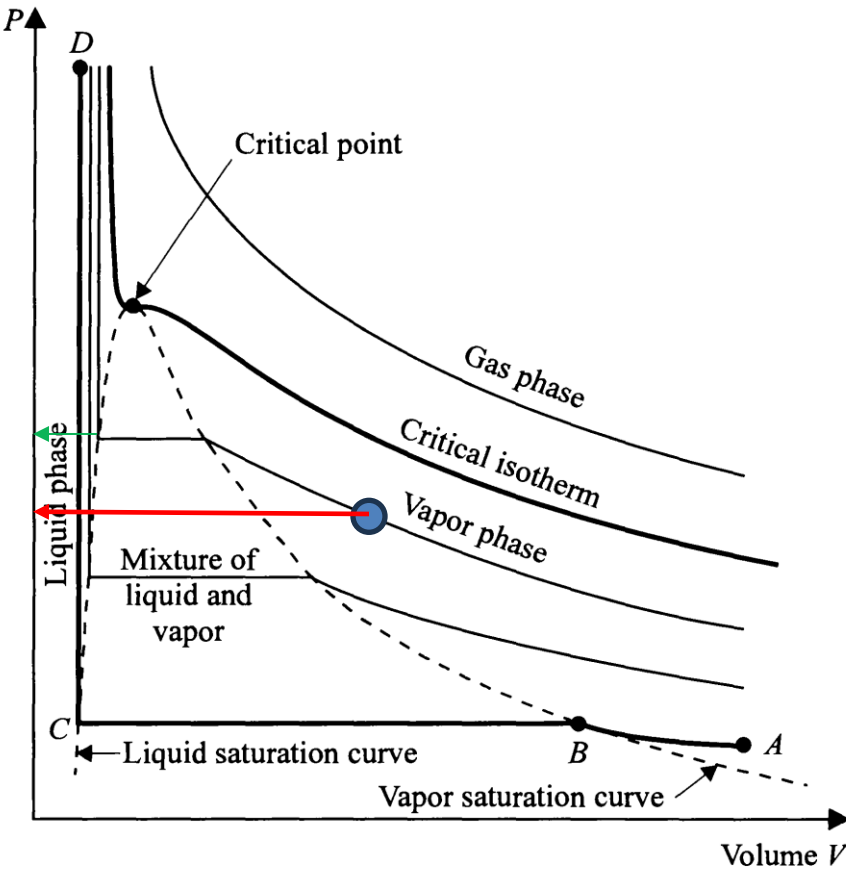
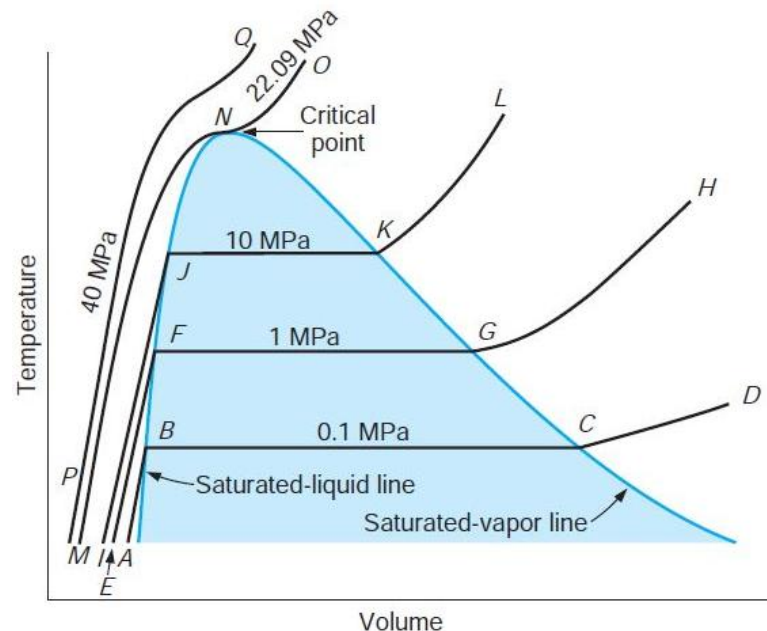


Fig. 2

Saturated Liquid, Saturated Vapor, Subcooled/compressed Liquid, Superheated vapor/Unsaturated Vapor



Now let the process take place at a constant pressure of 1 MPa, starting from an initial temperature of 20°C. Point E represents the initial state, in which the specific volume is slightly less than that at 0.1 MPa and 20°C. Vaporization begins at point F, where the temperature is 179.9°C. Point G is the saturated-vapor state, and line GH is the constant pressure process in which the steam is superheated. In a similar manner, a constant pressure of 10 MPa is represented by line IJKL, for which the saturation temperature is 311.1°C. At a pressure of 22.09 MPa, represented by line MNO, we find, however, that there is no constant temperature vaporization process. Instead, point N is a point of inflection with a zero slope and zero curvature. This point is called the critical point. At the critical point the saturated-liquid and saturated-vapor states are identical. The temperature, pressure, and specific volume at the critical point are called the critical temperature, critical pressure, and critical volume.



A constant-pressure process at a pressure greater than the critical pressure is represented by line PQ . If water at **40 MPa** and **20°C** is heated in a constant-pressure process in a cylinder, as shown in Fig. 1, **two phases will never be present and the state shown in Fig. 1b will never exist**. Instead, there will be a continuous change in density, and at all times only one phase will be present. The question then is, when do we have a liquid and when do we have a vapor? The answer is that this is not a valid question at supercritical pressures. We simply call the substance a fluid. However, rather arbitrarily, at temperatures below the critical temperature we usually refer to it as a compressed liquid and at temperatures above the critical temperature as a superheated vapor. It should be emphasized, however, that at pressures above the critical pressure a liquid phase and a vapor phase of a pure substance never exist in equilibrium.

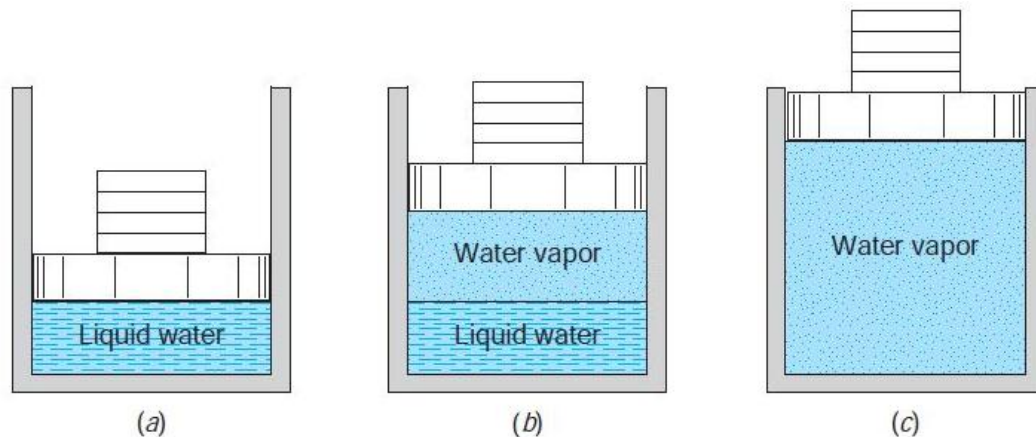


Fig.1

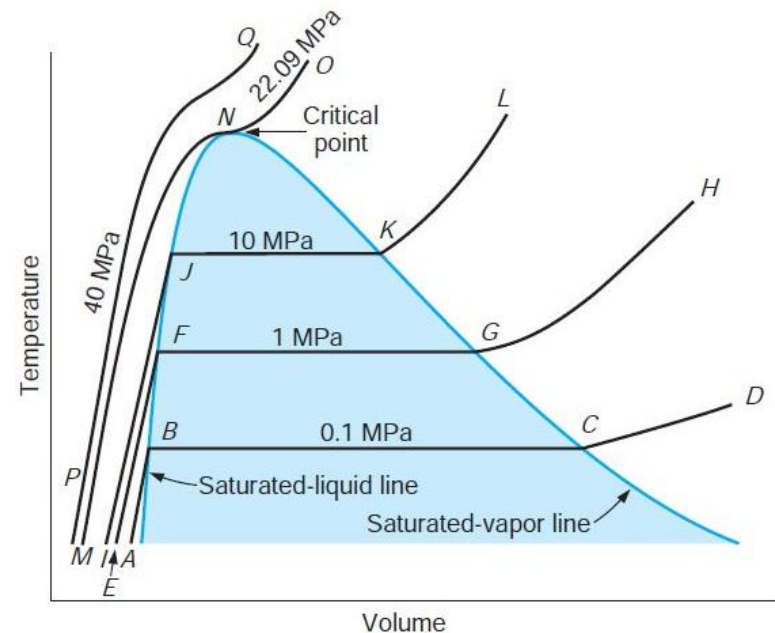
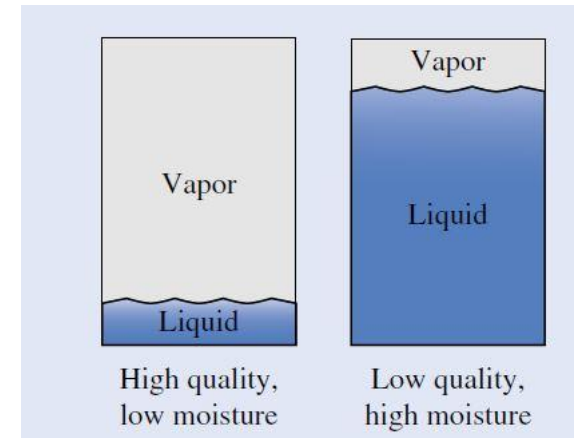


Fig.3

Why do they call it quality (x)

In the 19th century, there were a lot of steam engines. Railroads, factories, ships, and so on all used steam engines as their source of power. The people responsible for keeping these engines running noticed that they worked better if the steam contained more vapor than liquid. So, a mixture that contained a lot of vapor and little liquid was said to be of high quality.



Robert Balmer

Saturated with what?

The term saturated comes from the 18th century, when heat was thought to be a fluid. At that time, it was thought that a substance could be saturated with heat, just like water can become saturated with salt or sugar. Today we recognize that heat is not a fluid, and therefore the use of the word saturation in reference to a thermodynamic phase change is really a misnomer. However, this term is now completely entrenched in modern thermodynamic literature and cannot be changed.

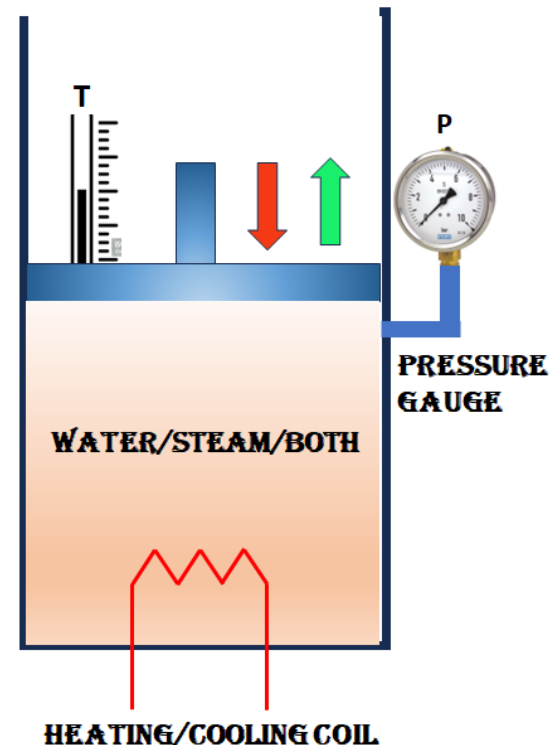
Saturated Liquid water and Steam

T	P	v_f	v_g	v_{fg}	u_f	u_g	u_{fg}	pv_f	pv_g	$u_f + pv_f$	$u_g + pv_g$
°C	(kPa)	m³/kg	m³/kg	m³/kg	kJ/kg	kJ/kAg	kJ/kg	kJ/kg	kJ/kg	kJ/kg	kJ/kg
0.01	0.6113	0.001000	206.131	206.131	0	2375.33	2375.33	0.000611	126.0079	0.000611	2501.338
5	0.8721	0.001000	147.117	147.118	20.97	2382.24	2361.27	0.000872	128.3007	20.97087	2510.541
100	101.3	0.001044	1.67185	1.67290	418.19	2506.50	2087.58	0.105757	169.3584	418.2958	2675.858
190	1254.4	0.001141	0.15654	0.15539	806.17	2590.01	1783.84	1.43127	196.3638	807.6013	2786.374
250	3973	0.001251	0.05013	0.04887	1080.37	2602.37	1522.00	4.970223	199.1665	1085.34	2801.536
300	8581	0.001404	0.02167	0.02027	1331.97	2562.96	1230.99	12.04772	185.9503	1344.018	2748.91
374.1	22089	0.003155	0.003155	0	2029.58	2029.58	0	69.6908	69.6908	2099.271	2099.271

T	T	P	h_f	h_g	h_{fg}	s_f	s_g	s_{fg}	$g_f = h_f - Ts_f$	$g_g = h_g - Ts_g$
°C	K	(kPa)	kJ/kg	kJ/kg	kJ/kg	kJ/kg-K	kJ/kg-K	kJ/kg-K	kJ/kg	kJ/kg
0.01	273.16	0.6113	0.00	2501.35	2501.35	0	9.1562	9.1562	0	0.242408
5	278.15	0.8721	20.98	2510.54	2489.57	0.0761	9.0257	8.9496	-0.18721	0.041545
100	373.15	101.3	419.02	2676.05	2257.03	1.3068	7.3548	6.0480	-68.6124	-68.3936
190	463.15	1254.4	807.61	2786.37	1978.76	2.2358	6.5078	4.2720	-227.901	-227.718
250	523.15	3973	1085.34	2801.52	1716.18	2.7927	6.0729	3.2802	-375.661	-375.518
300	573.15	8581	1344.01	2748.94	1404.93	3.2533	5.7044	2.4511	-520.619	-520.537
374.1	647.25	22089	2099.26	2099.26	0	4.4297	4.4297	0	-767.863	-767.863

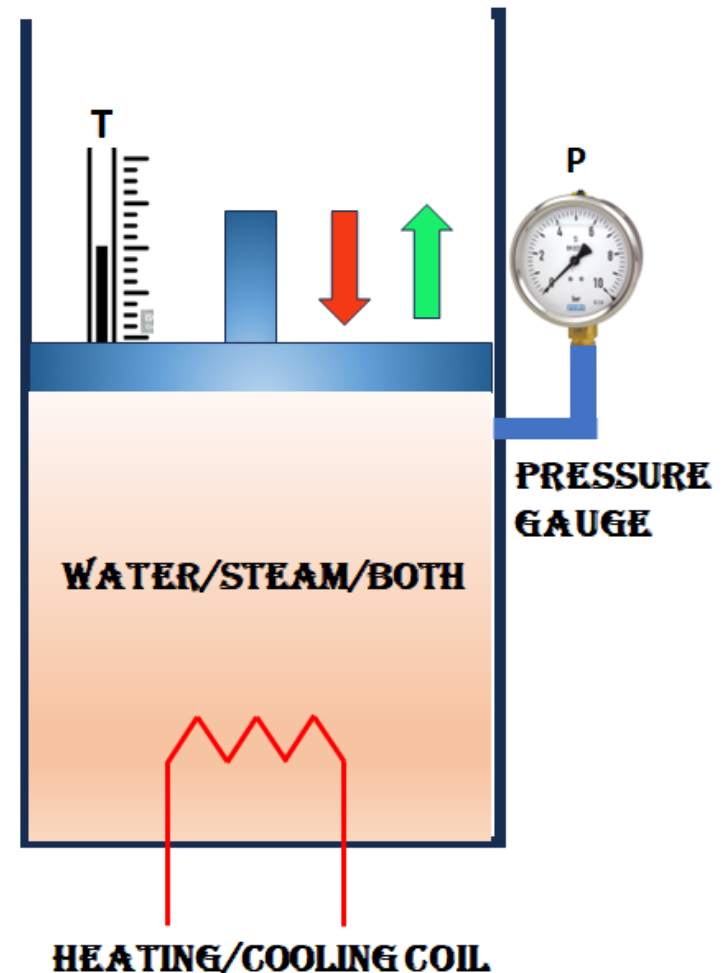
P-V diagram for a pure substance

- Since the triple point of water is the basis for the thermodynamic temperature scale, let us investigate pure water as an example of a pure substance.
- The experiment consists of measurement of P and V in states of equilibrium for H_2O either as a single phase or coexisting as multiple phases.
- The sample is placed in a cylinder closed at one end and contained by a piston. No mass is allowed to escape. Ovens and refrigerators control the temperature of the system. Movement of the piston provides independent control of the volume of the system. Pressure is measured by means of a pressure gauge. A small window permits observing the state of the sample.

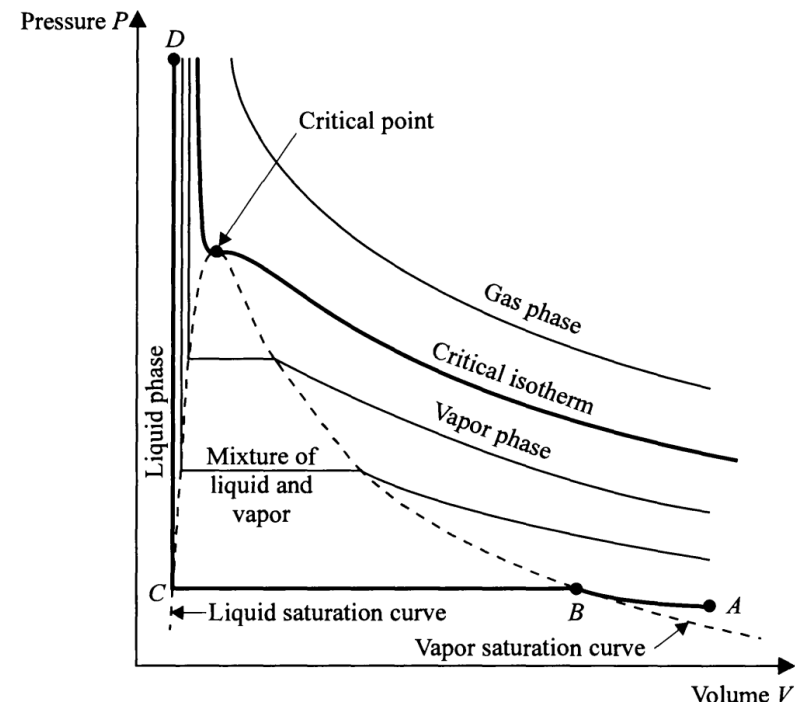
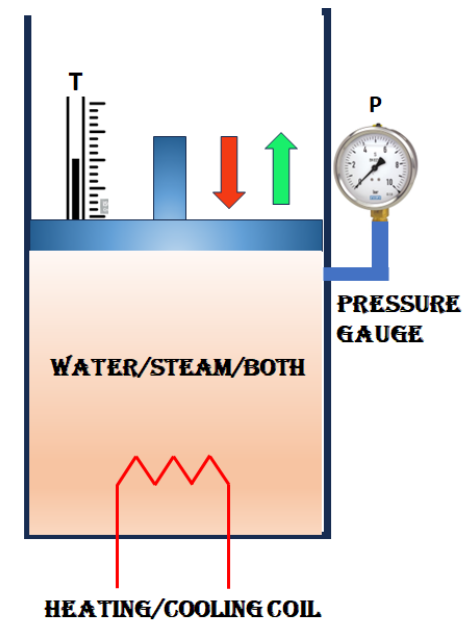


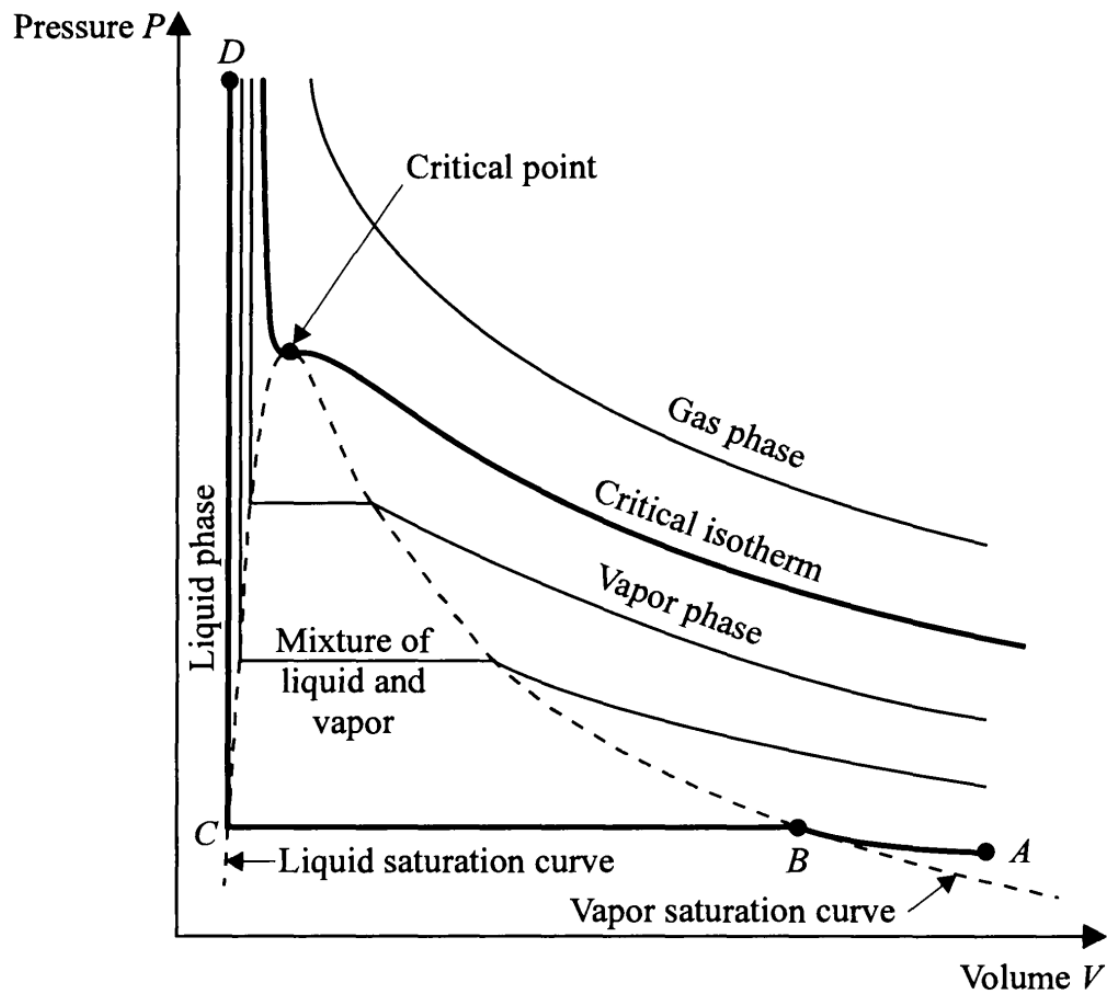
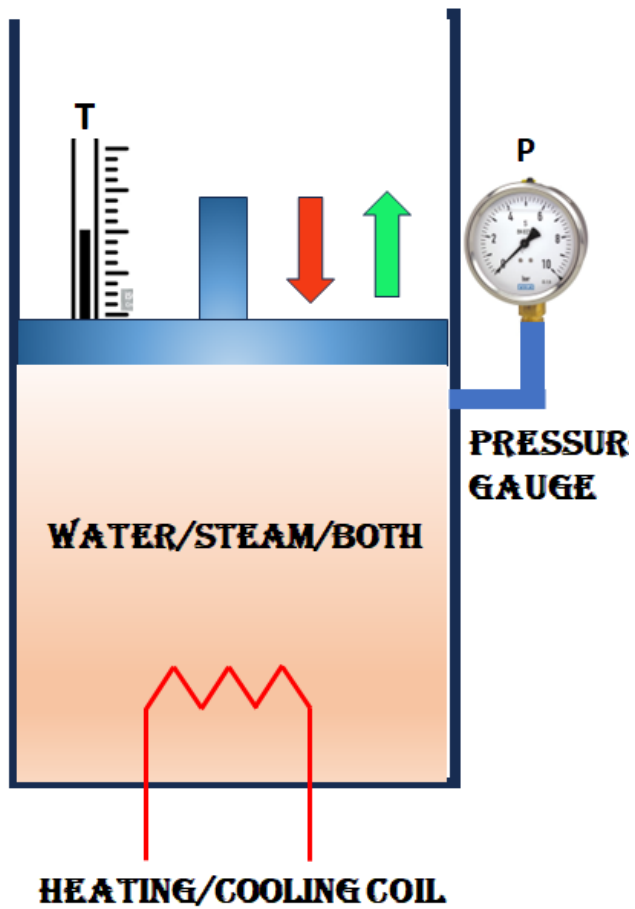
P-V diagram for a pure substance

If 1 g of water at 94°C ($P_{\text{sat}} = 81.47 \text{ kPa}$) is introduced into the cylinder 2 liters in volume, from which **all air has been removed**, **the water will evaporate completely** and the system will be in the condition of **unsaturated vapor**, the pressure of the vapor being less than standard atmospheric pressure



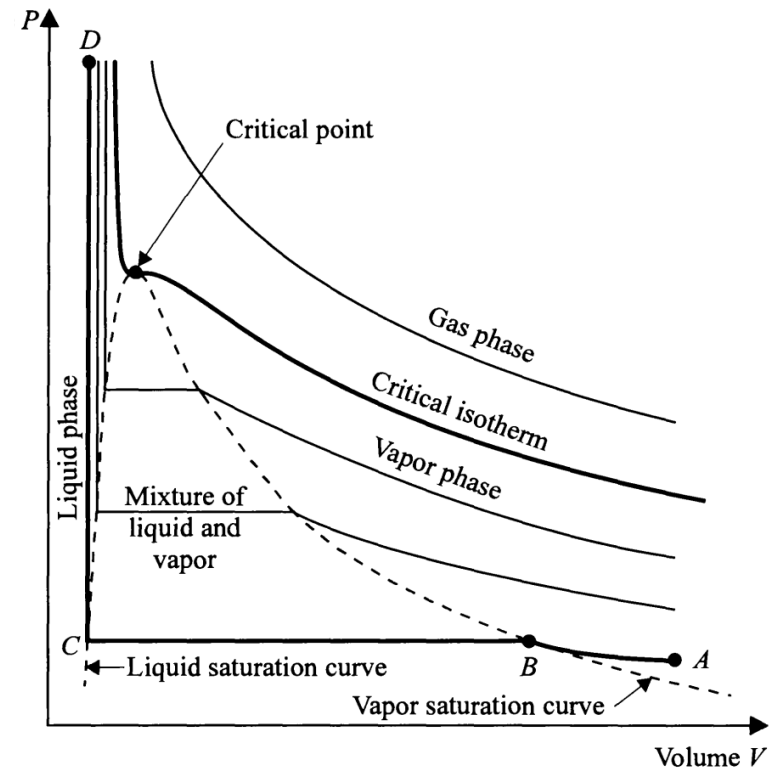
On P-v diagram shown in the figure, this state is represented by **point A**. If the vapor is then **compressed slowly and isothermally**, obviously the **volume will be decreasing**, but more importantly the **pressure will rise until the system reaches a state of saturated vapor** represented by **point B** on the isotherm. If the compression is continued, condensation occurs to form droplets of water. During the compression the pressure remains constant (**isobaric process**) as long as the temperature remains constant. The **horizontal straight-line segment BC** represents the **isothermal and isobaric condensation of water vapor** as the volume decreases, the constant pressure being the **vapor pressure**. Alternatively, as the volume increases, **the vaporization line BC** represents the **isothermal isobaric evaporation of water vapor**. At any point between B and C, water and vapor coexists in equilibrium. **At point C the sample is only liquid water, or saturated liquid**.



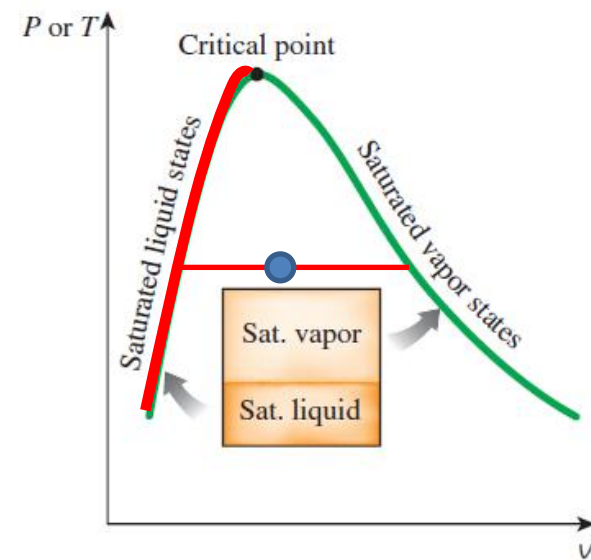


In order to compress liquid water slightly very large increase in pressure is required hence the line CD is almost vertical. At any point on CD, the water is said to be in the liquid phase; at any point on AB in the vapor phase; at any point on BC between the two saturated curves, there is equilibrium between liquid and vapor phases. The curves ABCD, which is discontinuous at point B and C is typical isotherm of a pure substance on a PV diagram

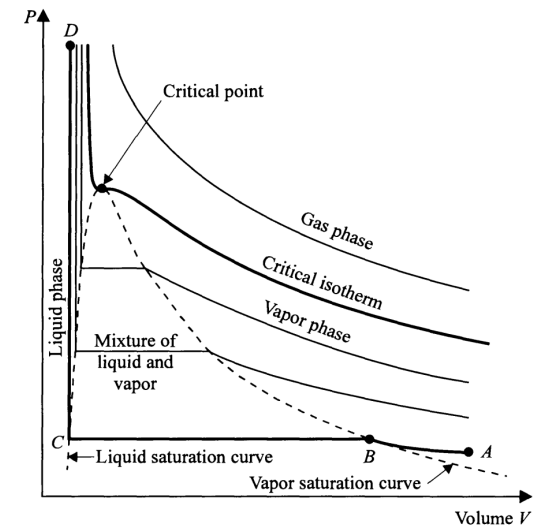
At other temperature the isotherm in the liquid vapor region have a similar character. It is seen that the vaporization lines representing equilibrium between coexisting liquid and vapor phases become shorter and vanishes at certain temperature called critical temperature. The isotherm at critical temperature is called critical isotherm and the point that represents the limit of vaporization lines is called critical point. It is seen that critical point is a point of inflection on the critical isotherm. The pressure and volume at the critical point are known as critical pressure and critical volume respectively.



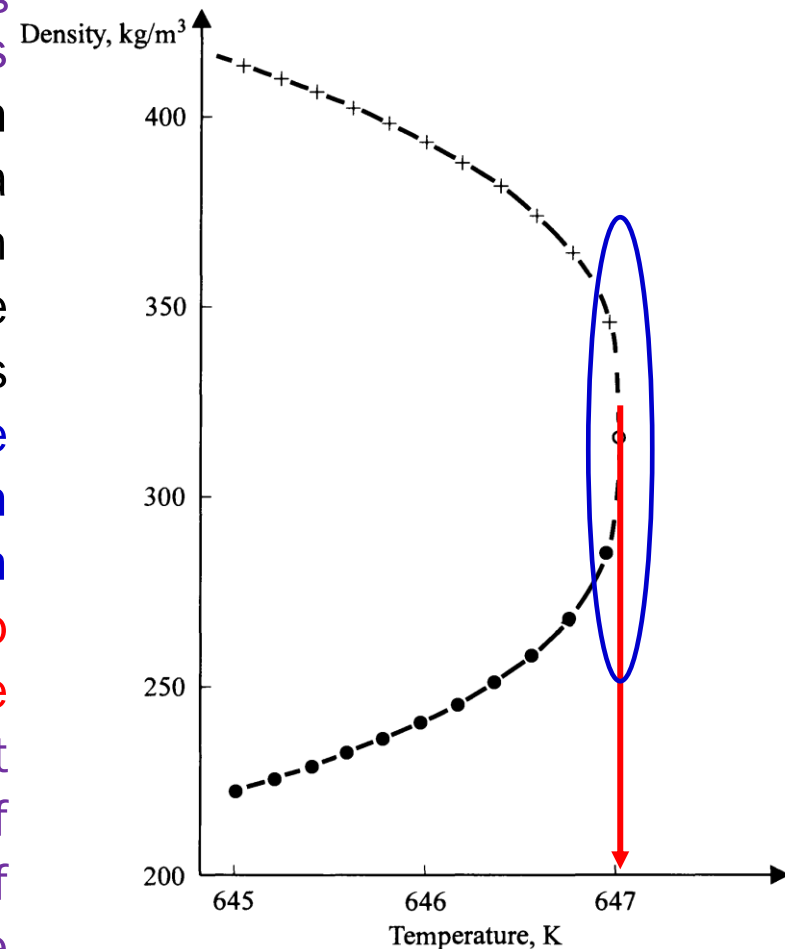
All point at which the liquid is saturated lie on liquid saturation curve, and all points representing saturated vapor lie on the vapor saturation curve. These two saturation lines denoted by the red and green lines meet at the critical point.



Above the critical point the isotherms are continuous curves, which at large volume and low pressure lose their inflection point and approach equilateral hyperbolas, namely the isotherm of an ideal gas. At temperature above the critical. Temperature, there is no longer any distinction between liquid and vapor, that is, meniscus between the liquid vapor disappear. For water the critical temperature is **647.067 K** and critical pressure is **22.046 MPa**, the critical volume for 1 kg is **0.00309 m³** and the critical density is **322.778 kg/m³**



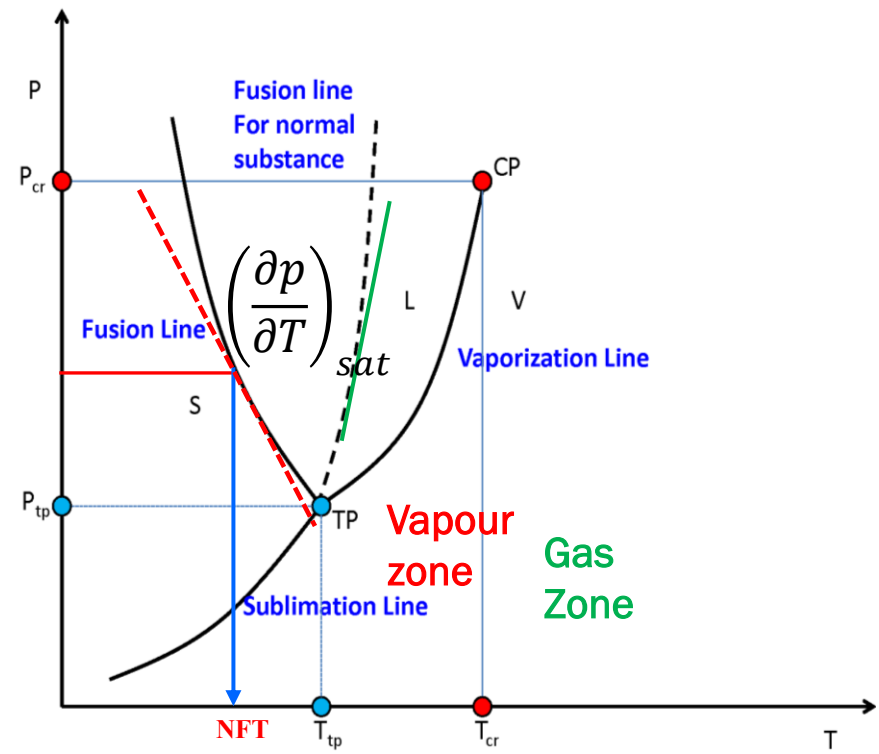
It is clear that the critical point is the limiting point at which the specific volume of the liquid is equal to that of the vapor. If the densities of both liquid and vapor of water are measured as a function of temperature and are plotted as shown in the figure, the critical temperature can be determined from the point where both the curves meet. At temperature below the critical point the liquid and vapor densities vary only slightly with temperature and are significantly different from each other. It is interesting that at no temperature near the critical temperature is the meniscus visible between the liquid and vapor. It was once thought that the disappearance of meniscus was the criterion for the attainment of the critical point. At temperature just below the critical point, there are two distinct phases with different densities but they cannot be distinguished visually. Although critical point represents a unique temperature, it is not easily achieved or maintained; therefore, it is not used in thermometry as a reference temperature.



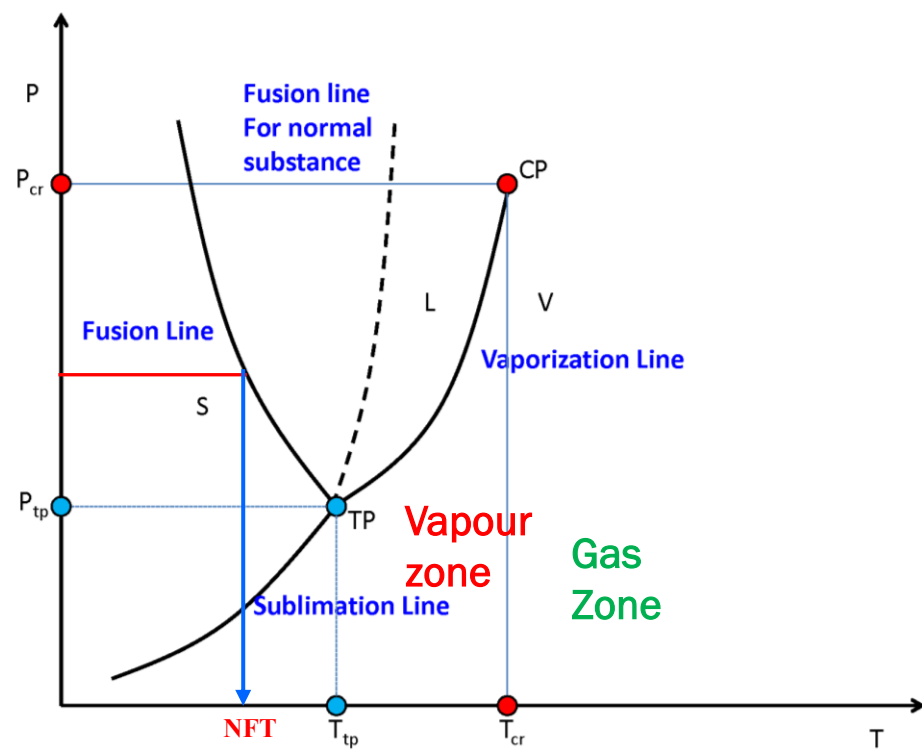
← Imp.

P-T diagram

The **P-T diagram** is otherwise known as the **phase diagram** which represents the various **coexistence curves** such as **fusion line**, **vaporization line** and **sublimation lines**. The slope of these lines given by the Clapeyron equation gives the rate of change of saturation pressure with temperature. **For all the substance the vaporization and sublimation line are having positive slope**. However, the negative slope of fusion line of water results in depression in freezing point with increase in pressure (**This is the reason why the NFT (Normal freezing temperature) of water is less than the triple point**).

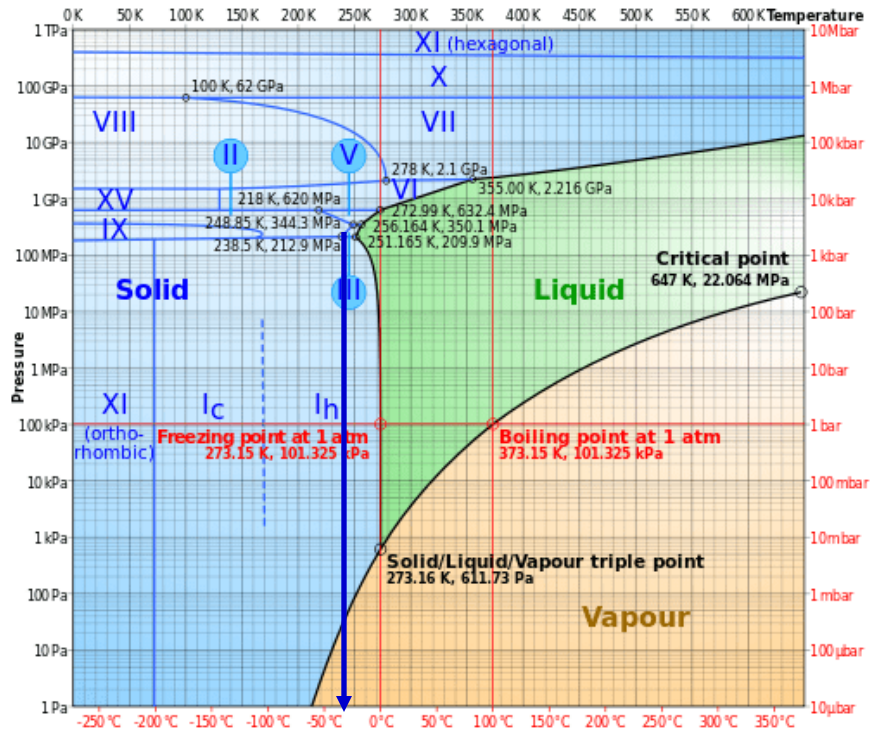


These three coexistence curves meet at the triple point. Critical point is one of the extreme end of the vaporization curve above which the liquid can not be in equilibrium with its vapor phase. It is the point discontinuity on the vapor dome. Abrupt change in the thermodynamic as well as transport properties are observed near the critical point. At CP the surface tension goes to zero, so the vapor and liquid phase are not distinguishable



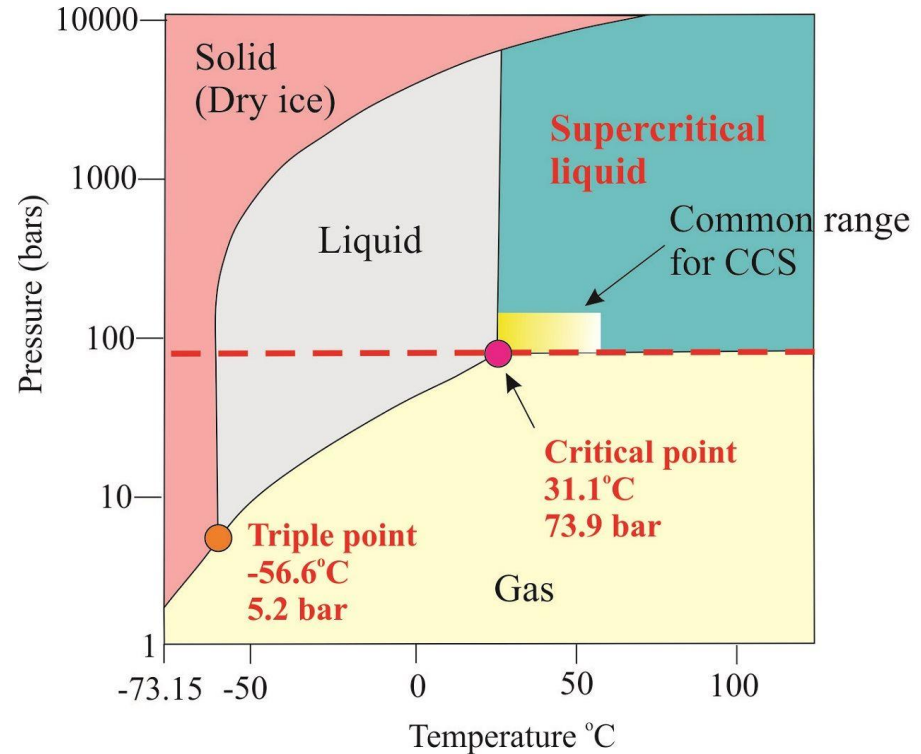
Substance	P_{cr} (bar)	T_{cr} (°C)	P_{tp} (bar)	T_{tp} (°C)	NBT (°C)	NFT (°C)
H ₂ O	221.2	374.3	0.006112	0.01	100	0
CO ₂	73.8	31	5.18	-56.6	-78.46	-----

P-T diagram (Phase Diagram of Water and CO₂)



Phase Diagram of Water

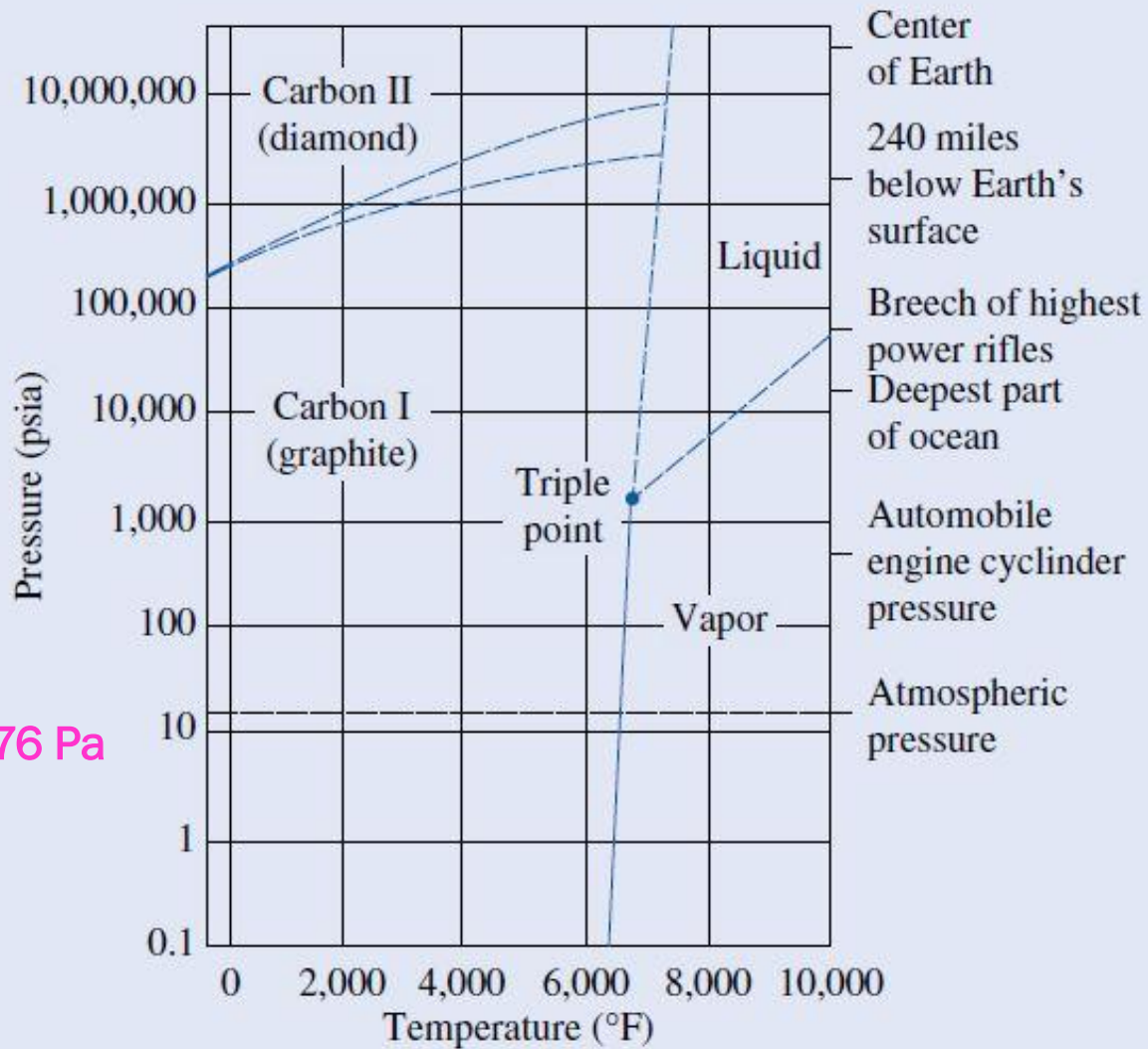
<https://www.google.com>



Phase Diagram of CO₂

<https://www.google.com>

Phase diagram of Diamond



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